



PROJECT REPORT



Abstract

Cybersecurity threats continue to increase in complexity and frequency, making continuous monitoring and threat detection essential for protecting organizational assets. This project presents the design and implementation of a Security Information and Event Management (SIEM) laboratory environment using the Wazuh platform. The implemented environment consists of an Ubuntu-based Wazuh server, a Windows 10 endpoint, and a Kali Linux attacker machine operating within an isolated virtualized network.

The project demonstrates centralized log collection, file integrity monitoring, endpoint security monitoring, attack simulation, custom rule development, and security event analysis. Real-world attack scenarios including SSH authentication attacks and Hydra-based brute force attacks were performed to evaluate the effectiveness of the monitoring infrastructure. Custom Wazuh detection rules were implemented to enhance threat visibility and improve incident detection capabilities. The results show that Wazuh successfully collected, analyzed, and correlated security events while generating meaningful alerts for suspicious activities.

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Chapter 1: Introduction

1.1 Background

Organizations face increasing cybersecurity risks from unauthorized access, malware infections, credential theft, and insider threats. Traditional security controls often operate independently, making it difficult to obtain a centralized view of security events occurring across multiple systems.

Security Information and Event Management (SIEM) solutions address this challenge by collecting logs from multiple devices, correlating events, and generating alerts when suspicious activities occur. Wazuh is an open-source SIEM and Extended Detection and Response (XDR) platform that provides centralized security monitoring, log analysis, intrusion detection, vulnerability assessment, file integrity monitoring, and incident response capabilities.

This project focuses on implementing a practical Wazuh-based cybersecurity monitoring environment capable of detecting and analyzing security events generated from multiple systems.

1.2 Project Objective

The primary objectives of this project are:

- To deploy a centralized SIEM infrastructure using Wazuh.
- To collect and analyze security logs from multiple endpoints.
- To implement File Integrity Monitoring (FIM) for detecting unauthorized file modifications.
- To perform endpoint security monitoring using Wazuh agents.
- To simulate cyberattacks and evaluate detection capabilities.
- To develop custom security rules for enhanced threat detection.
- To generate security alerts and support incident analysis.

1.3 Features of the Project

The implemented system provides the following features:

- Centralized log collection and analysis.
- Real-time security monitoring.
- File Integrity Monitoring (FIM).

- Endpoint Detection and Response (EDR).
- SSH authentication monitoring.
- Brute force attack detection.
- Custom threat detection rules.
- Security alert generation.
- Security event correlation.
- Dashboard-based visualization.
- MITRE ATT&CK mapping.
- Security incident investigation support.

1.4 Project Scope

The scope of this project includes:

- Enterprise security monitoring environments.
- Security Operations Centers (SOCs).
- Educational cybersecurity laboratories.
- Threat detection and incident response training.
- Network and system security monitoring.
- Compliance monitoring and auditing.
- Research and development of SIEM technologies.

The implemented architecture can be expanded to support additional endpoints, servers, cloud environments, and advanced threat detection use cases.

Chapter 2: Tools and Technologies

2.1 Hardware Requirements

- Host Computer
- Minimum 16 GB RAM
- Multi-core Processor
- 250 GB Storage

2.2 Software Requirements

VMware Workstation

Used to create and manage isolated virtual machines.

Ubuntu Linux

Used as the Wazuh SIEM server.

Windows 10

Used as a monitored endpoint system.

Kali Linux

Used as an attacker machine for security testing.

Wazuh Platform

Used for centralized security monitoring and event management.

Nmap

Used for network reconnaissance and service discovery.

Hydra

Used for password attack simulation and brute-force testing.

Chapter 3: System Architecture

3.1 Architecture Overview

The implemented environment consists of three virtual machines:

Wazuh Server (Ubuntu)

Functions:

- Wazuh Manager
- Wazuh Indexer

- Wazuh Dashboard
- Event Correlation Engine

Windows Endpoint

Functions:

- Endpoint Monitoring
- Security Log Generation
- Wazuh Agent

Kali Linux

Functions:

- Attack Simulation
- Network Scanning
- Brute Force Testing

3.2 Communication Flow

1. Endpoints generate security events.
2. Wazuh agents collect logs.
3. Logs are forwarded to the Wazuh Manager.
4. Wazuh processes and correlates events.
5. Alerts are generated based on detection rules.
6. Security analysts investigate alerts through the dashboard.

Chapter 4: Log Collection Implementation

Overview

Centralized log collection was implemented using Wazuh agents deployed on monitored systems.

Ubuntu Log Sources

- Authentication Logs

- System Logs
- SSH Logs

Windows Log Sources

- Security Event Logs
- System Event Logs
- Application Logs

Outcome

Security events from multiple systems were successfully collected and processed through a centralized monitoring platform.

Chapter 5: File Integrity Monitoring

Overview

File Integrity Monitoring was configured to detect unauthorized modifications to critical files and directories.

Purpose

The purpose of FIM is to identify:

- File creation
- File deletion
- File modification
- Permission changes
- Unauthorized access attempts

Results

Wazuh successfully detected file changes and generated alerts whenever monitored files were modified.

This demonstrated the effectiveness of integrity monitoring controls within the environment.

Chapter 6: Endpoint Security Monitoring

Overview

Endpoint monitoring was performed using Wazuh agents installed on Windows and Ubuntu systems.

Security Events Monitored

- User authentication activities
- Failed login attempts
- Successful logins
- Privilege escalation events
- Process execution events
- File system modifications

Results

The endpoint monitoring infrastructure successfully collected and analyzed security-relevant events in real time.

Chapter 7: Attack Simulation and Detection

7.1 SSH Authentication Attack

An SSH authentication scenario was executed from Kali Linux against the Ubuntu target system.

Activities Performed

- SSH service verification
- Network connectivity testing
- Failed authentication attempt
- Successful authentication attempt

Detection Results

Wazuh generated alerts for:

- Failed SSH authentication
- Successful SSH authentication

- User session creation
- Authentication-related events

MITRE ATT&CK mappings were automatically associated with relevant events.

7.2 Hydra Brute Force Attack

A password brute-force attack was simulated using Hydra.

Activities Performed

- User enumeration preparation
- Password dictionary creation
- SSH brute-force attack execution
- Event monitoring

Detection Results

Wazuh successfully identified multiple failed authentication attempts generated during the attack simulation.

The attack activity was recorded within centralized security logs for investigation.

Chapter 8: Custom Detection Rules

To improve detection capabilities, custom Wazuh rules were developed.

Rule 1: SSH Login Failure Detection

Purpose:

Detect failed SSH authentication attempts.

Generated Alert:

Custom Alert: SSH Login Failure Detected

Severity Level:

Rule 2: SSH Brute Force Detection

Purpose:

Detect multiple failed SSH login attempts within a defined timeframe.

Generated Alert:

Custom Alert: Possible SSH Brute Force Attack

Severity Level:

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Benefits of Custom Rules

- Faster threat identification
- Improved visibility
- Reduced analyst workload
- Enhanced attack detection accuracy
- Better incident response support

Chapter 9: Security Event Analysis

Security events generated during attack simulations were analyzed through Wazuh logs and dashboard components.

Analysis Parameters

- Source IP Address
- Destination User
- Authentication Status
- Attack Type
- Event Severity
- MITRE ATT&CK Mapping

Findings

The monitoring platform successfully identified both legitimate and malicious activities while providing detailed forensic information for investigation.

Chapter 10: Results

The project successfully achieved all planned objectives.

Log Collection

Successfully implemented.

File Integrity Monitoring

Successfully implemented.

Endpoint Security Monitoring

Successfully implemented.

SSH Attack Detection

Successfully detected.

Hydra Brute Force Detection

Successfully detected.

Custom Rule Implementation

Successfully implemented.

Security Alert Generation

Successfully implemented.

Event Correlation and Analysis

Successfully implemented.

Overall, the system demonstrated effective real-time monitoring and threat detection capabilities.

Chapter 11: Conclusion

This project successfully implemented a comprehensive SIEM-based cybersecurity monitoring environment using Wazuh. The solution integrated centralized log collection, file integrity monitoring, endpoint security monitoring, attack detection, and custom rule development within a virtualized laboratory environment.

The implemented platform effectively detected SSH authentication attacks and brute-force activities while providing detailed security insights through centralized monitoring. The project demonstrates how open-source SIEM technologies can be utilized to improve organizational visibility, threat detection, and incident response capabilities.

Chapter 12: Future Work

Future improvements may include:

- Malware detection integration.
- Active response automation.
- Email alerting mechanisms.
- Threat intelligence integration.
- Cloud security monitoring.
- Vulnerability management integration.
- Machine learning-based anomaly detection.
- Advanced incident response automation.

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